FlowchartFun

Draw shapes

OCR GCSE Computer Science (J277) flowchart practice.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.

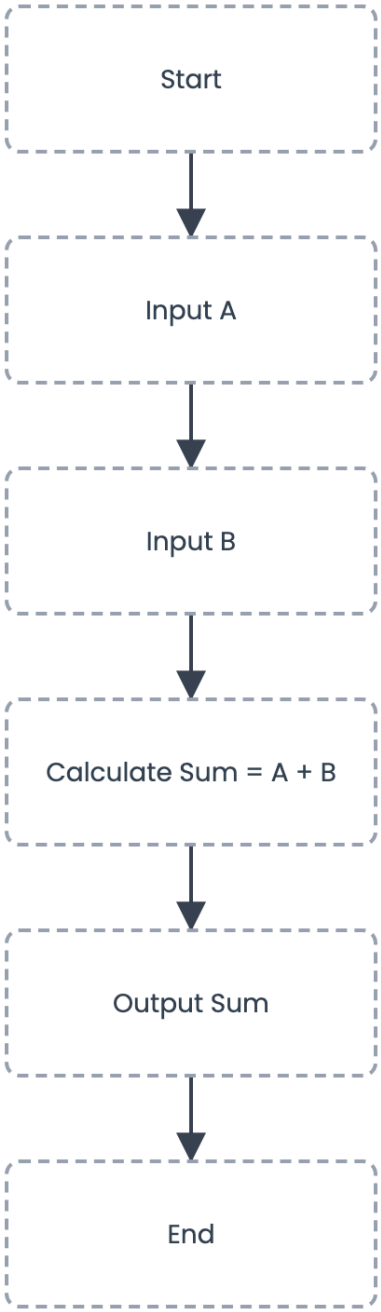
Name: _____ Class: _____

Level 1: Flowchart Basics

Easy · Draw Shapes

Create a simple flowchart that calculates the sum of two numbers. Drag each symbol into the correct place.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



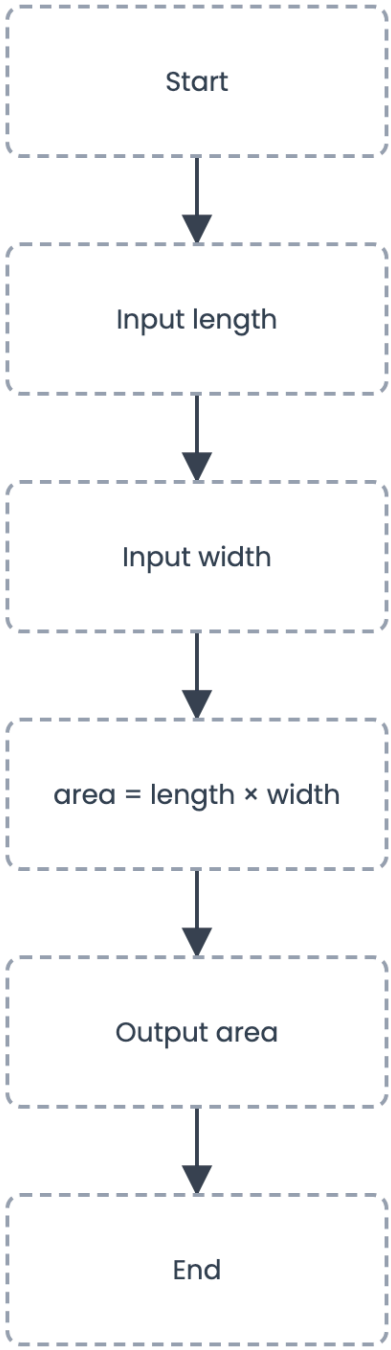


Level 2: Input, Process, Output

Easy · Draw Shapes

Make a flowchart that finds the area of a rectangle. Watch out: one symbol is not needed!

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



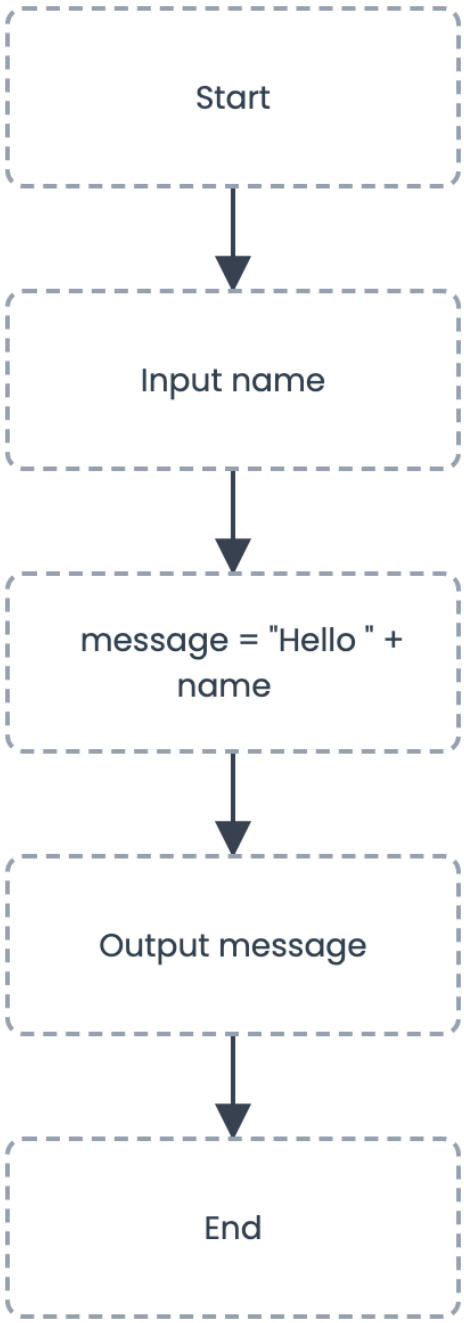


Level 3: Hello, Name

Easy · Draw Shapes

Ask for a name and greet the user. Not every symbol is needed.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



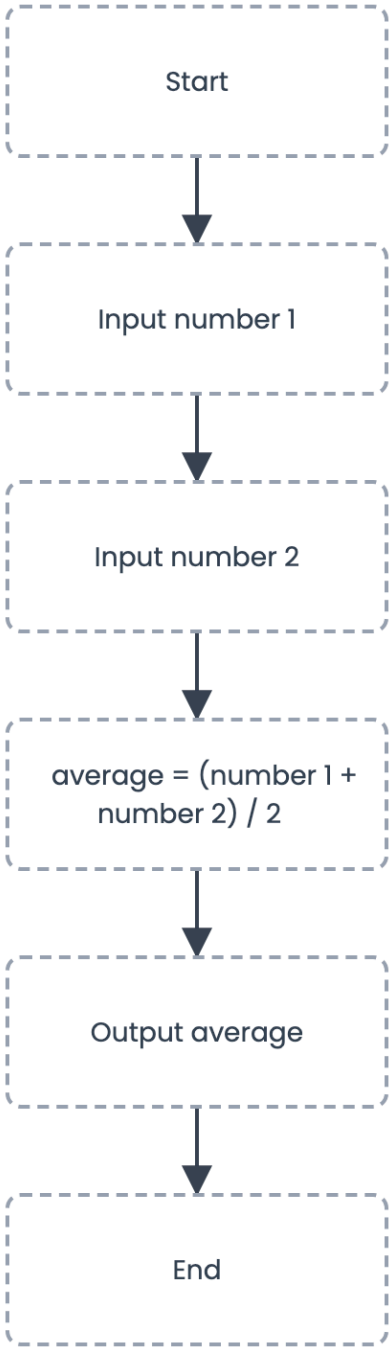


Level 4: Average of Two

Easy · Draw Shapes

Input two numbers and output their average. Remember the brackets!

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



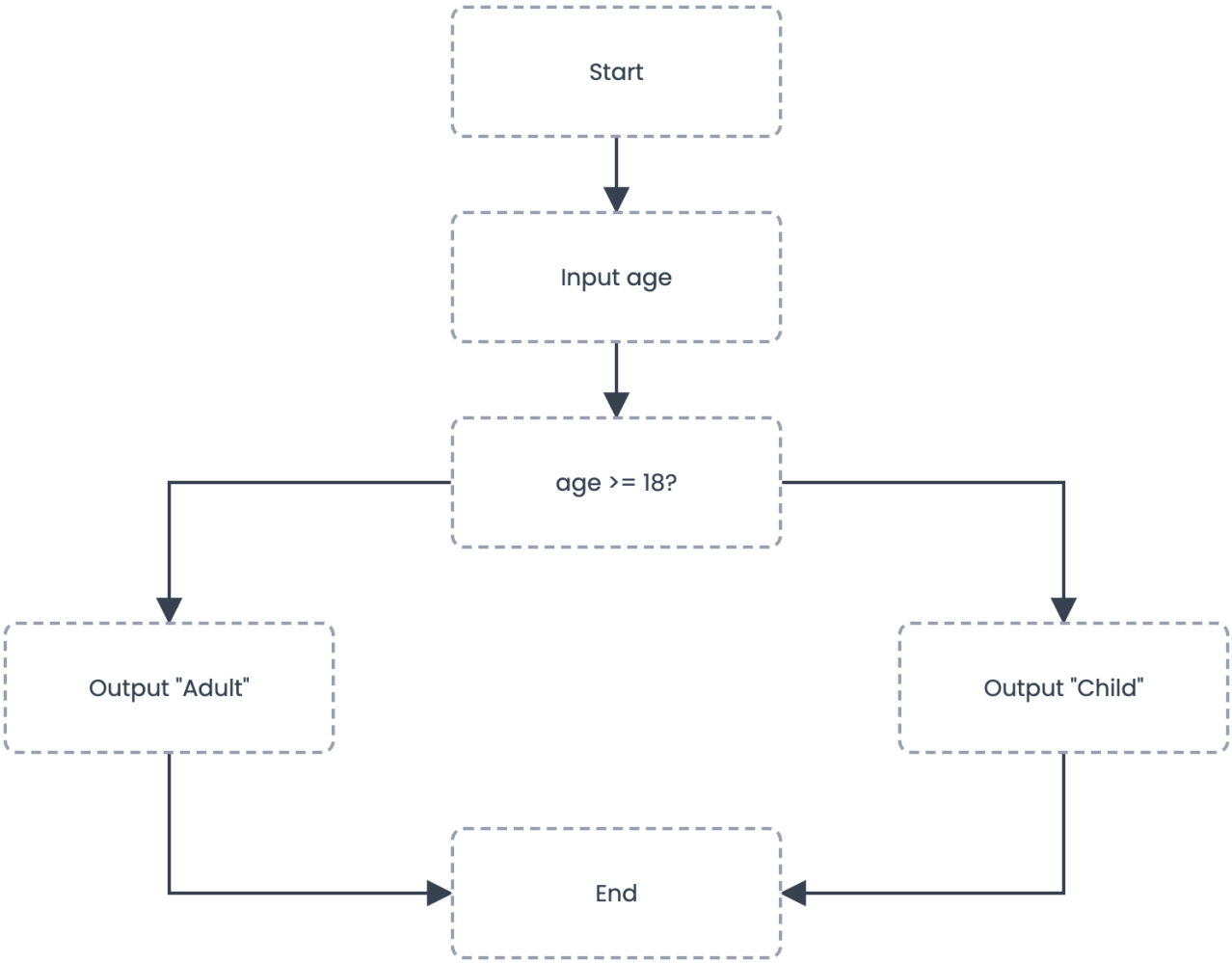


Level 5: Making a Decision

Easy · Draw Shapes

Output whether a person is an adult or a child. Remember: diamonds are decisions, with a Yes path and a No path.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.





Level 6: Odd or Even

Easy · Draw Shapes

Use MOD (the remainder after division) to decide whether a number is even or odd.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



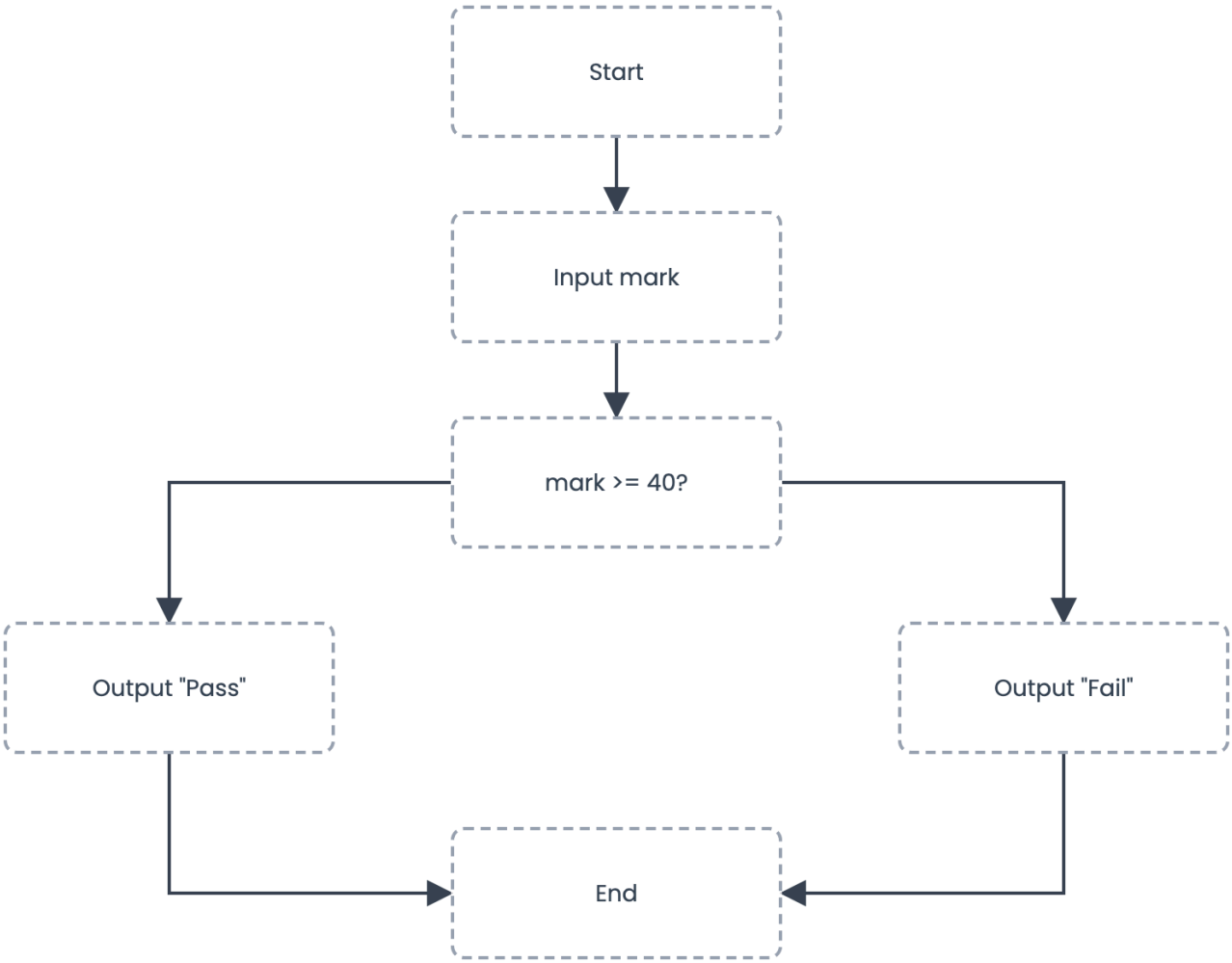


Level 7: Pass or Fail

Easy · Draw Shapes

A mark of 40 or more is a pass. Anything else is a fail.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



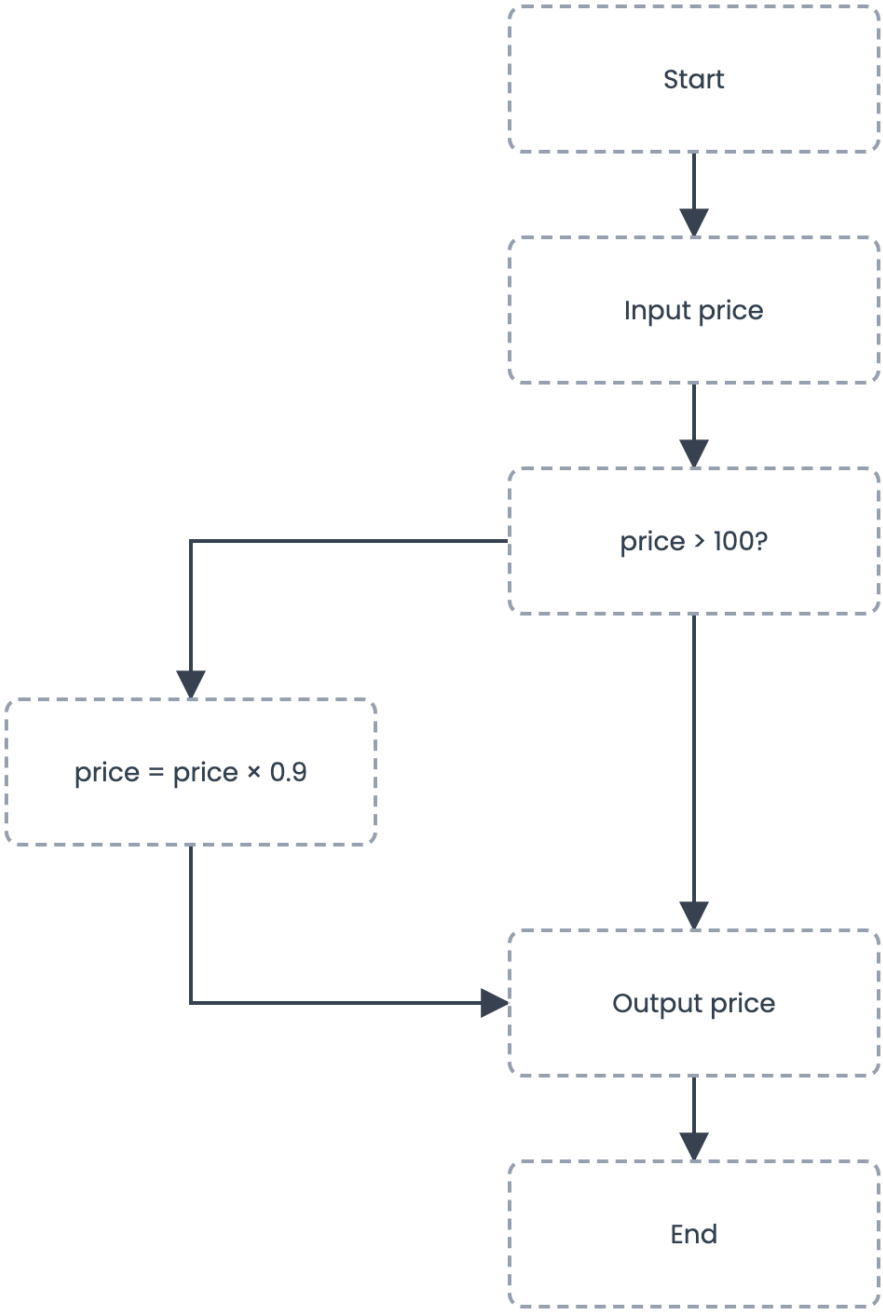


Level 8: Discount

Easy · Draw Shapes

Shoppers get 10% off if they spend more than £100. Everyone sees a price at the end. Notice that one path skips a step.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



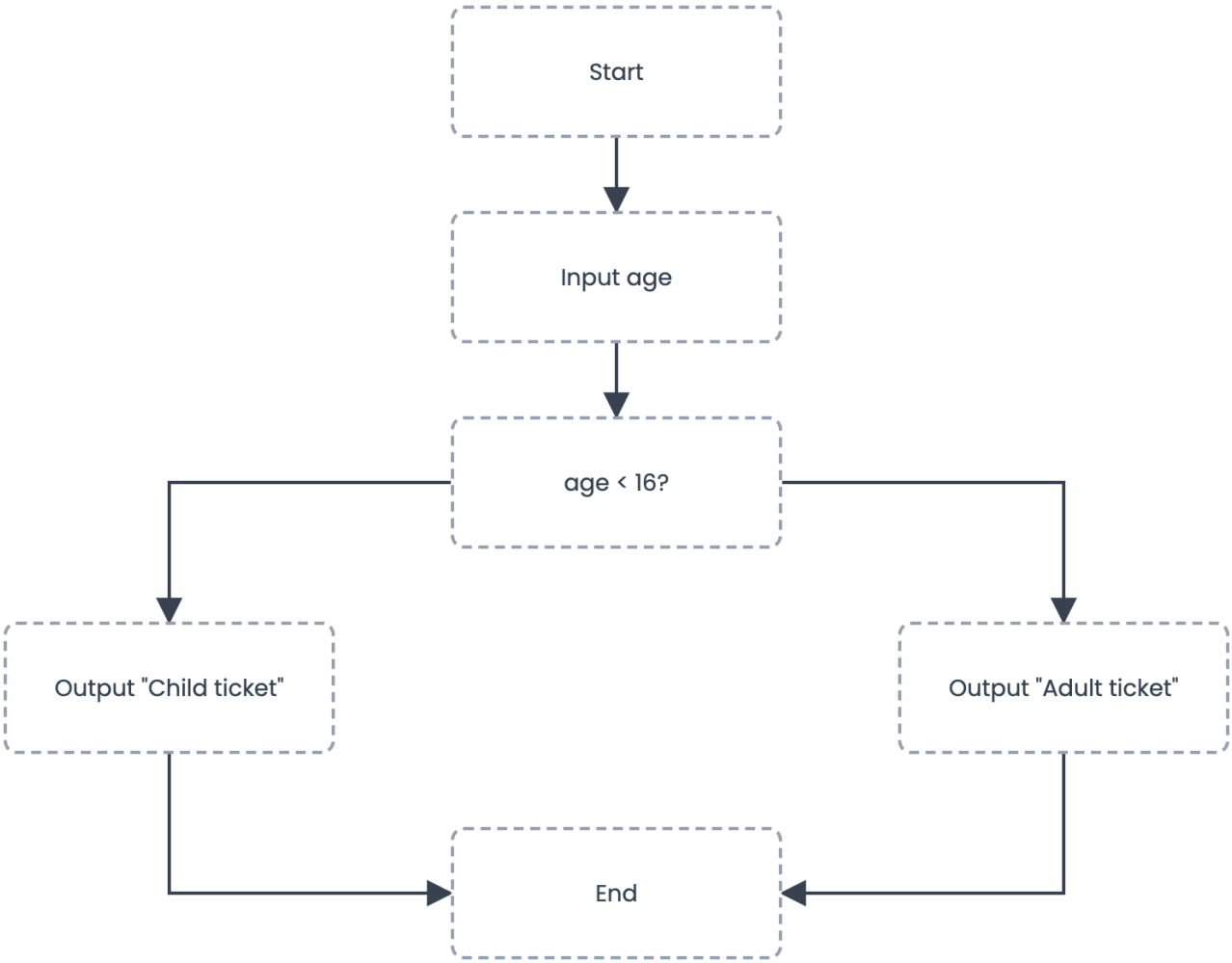


Level 9: Cinema Ticket

Easy · Draw Shapes

Under 16s pay the child price. Everyone else pays the adult price.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



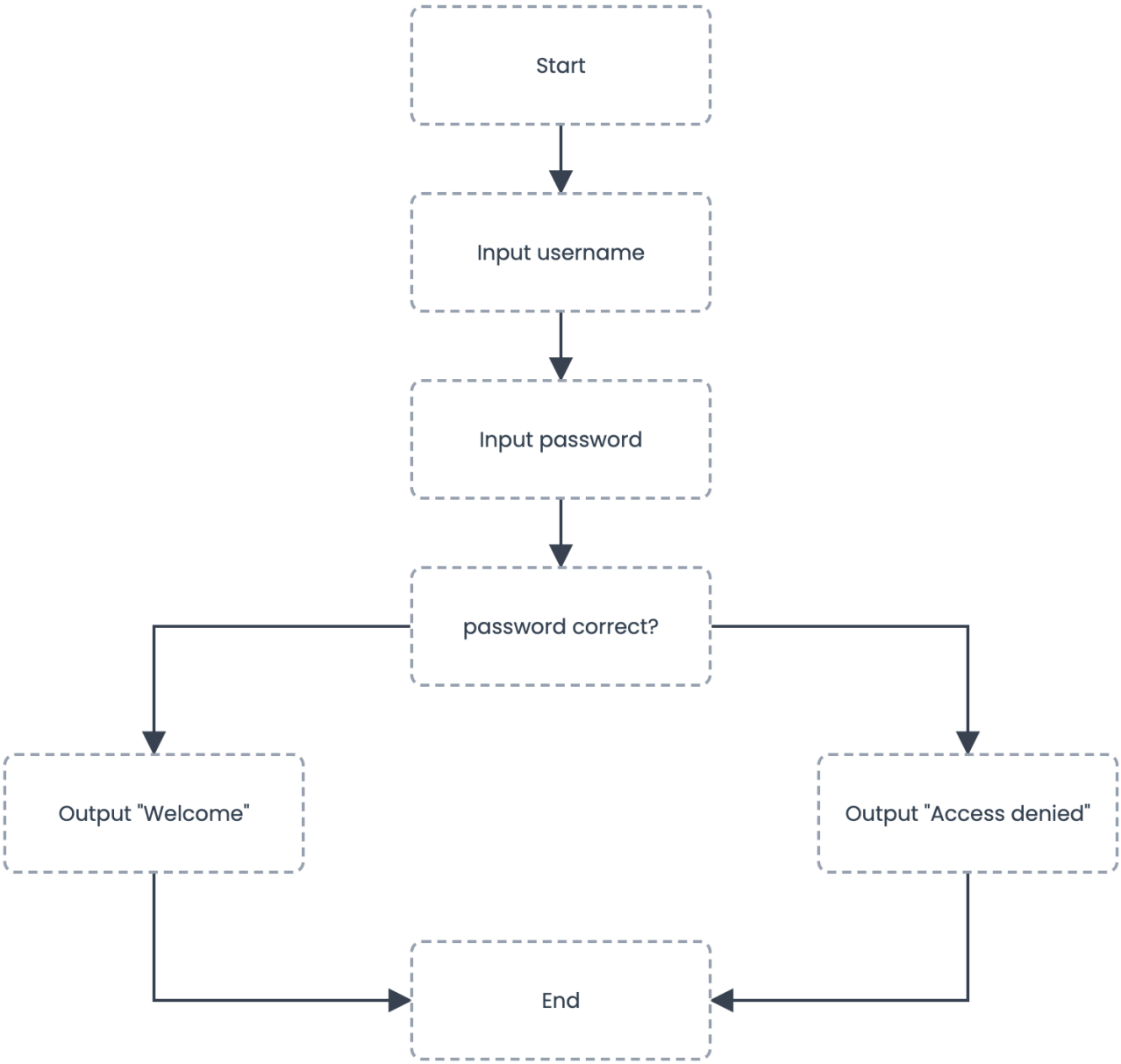


Level 10: Login

Easy · Draw Shapes

Ask for a username and password, then decide whether to let the user in.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



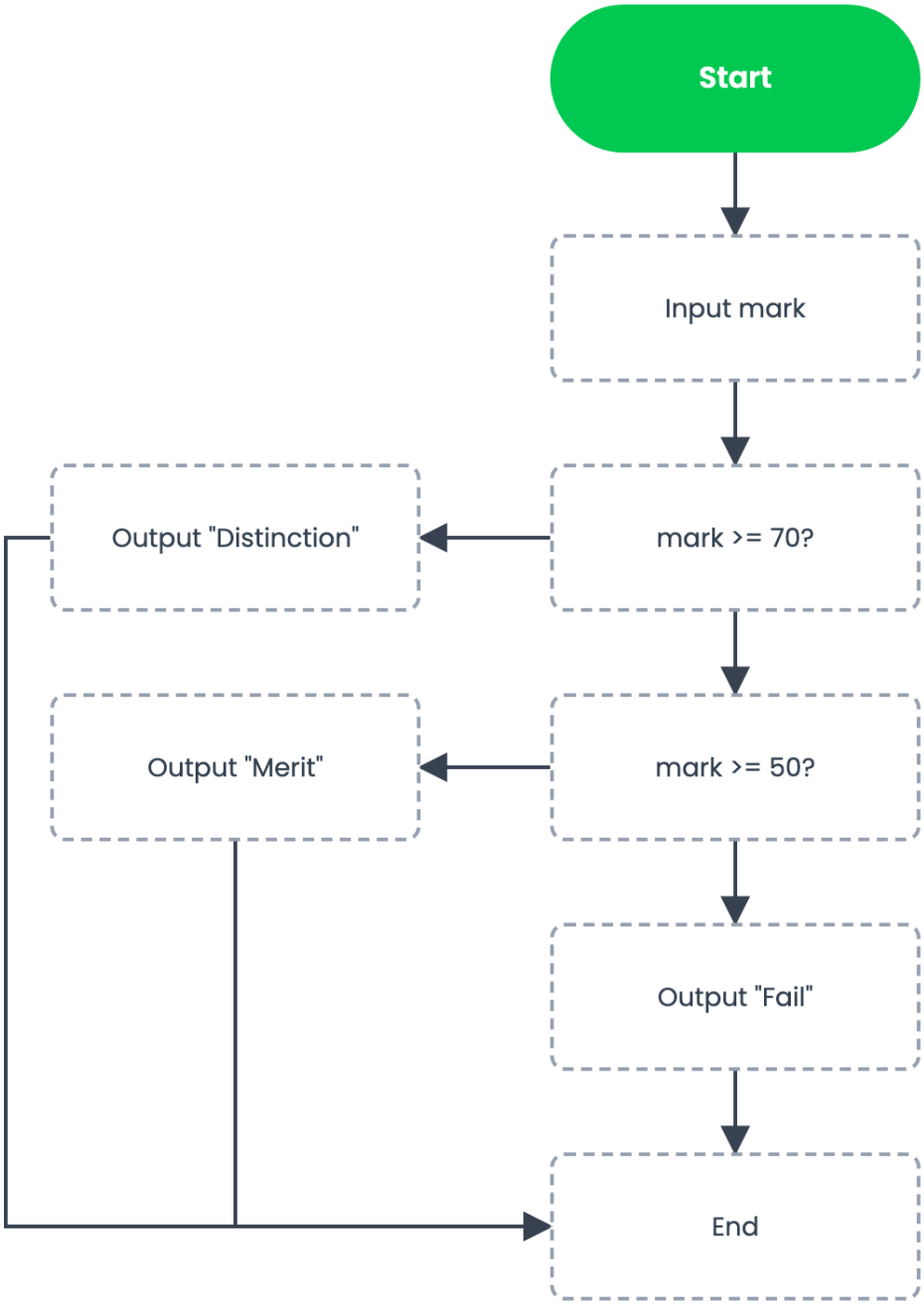


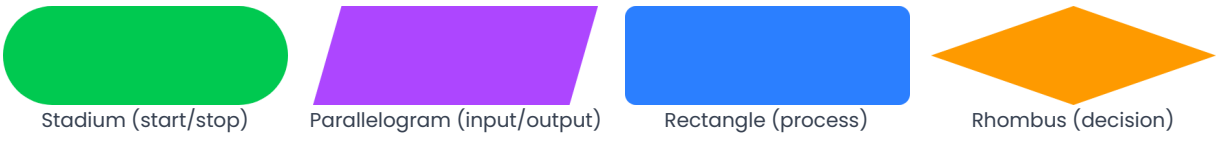
Level 11: Grade Boundaries

Medium · Draw Shapes

Output Distinction for 70 or more, Merit for 50 or more, otherwise Fail. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



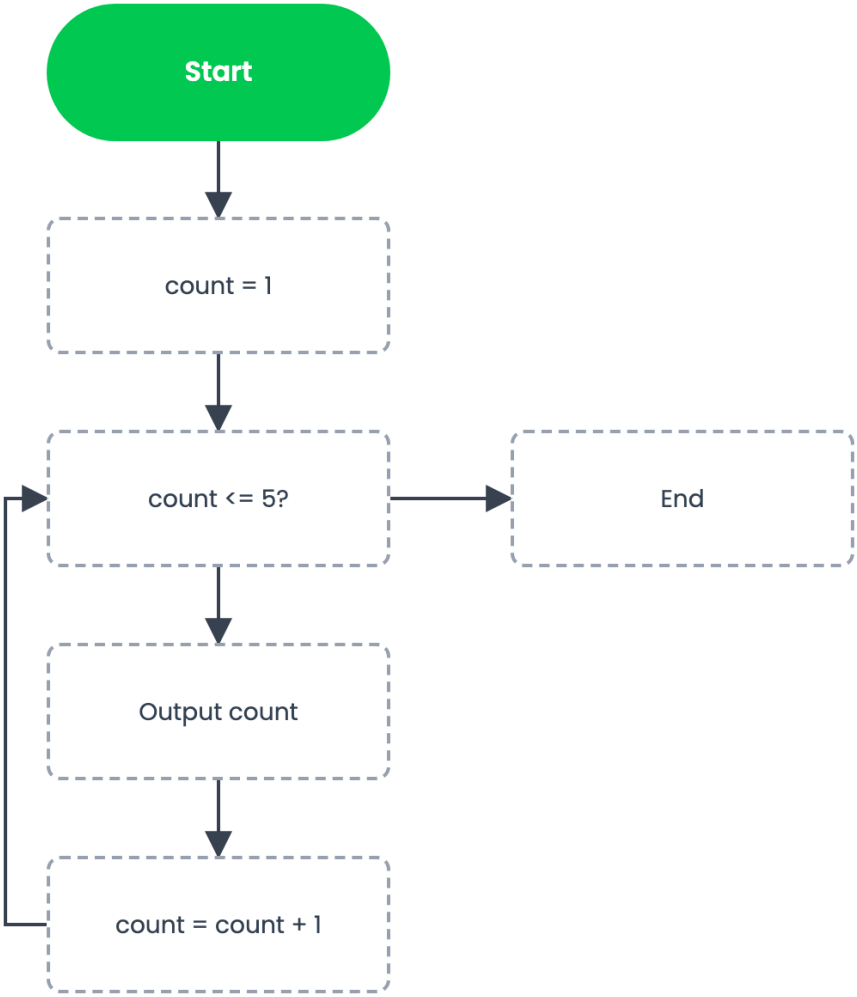


Level 12: Count to Five

Medium · Draw Shapes

Output the numbers 1 to 5. The arrow that goes back up is a loop. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



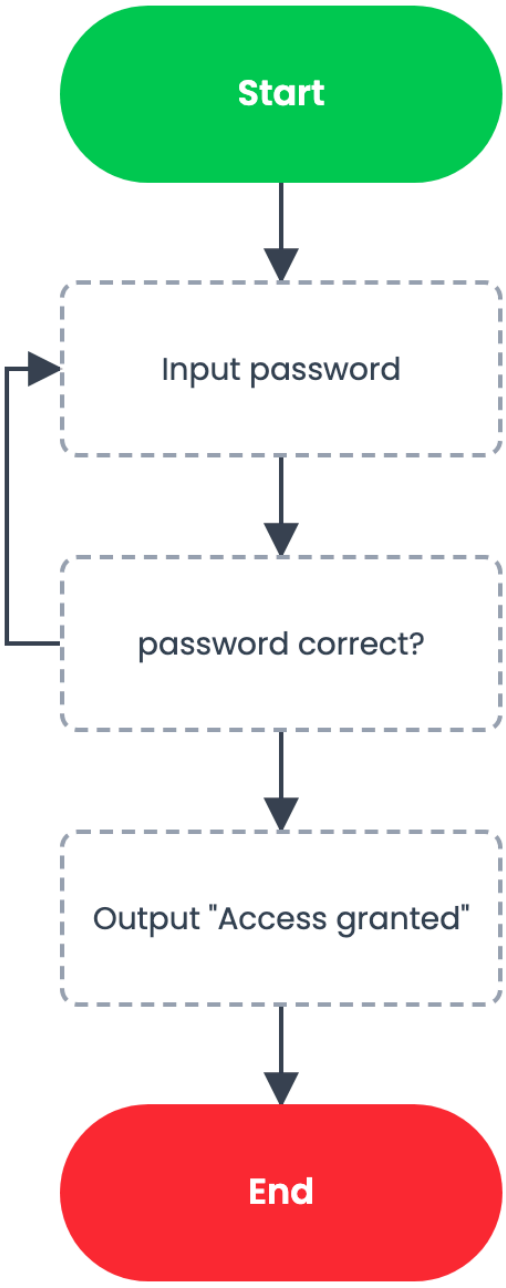


Level 13: Password Check

Medium · Draw Shapes

Keep asking for the password until it is correct, then grant access. Start and End are placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



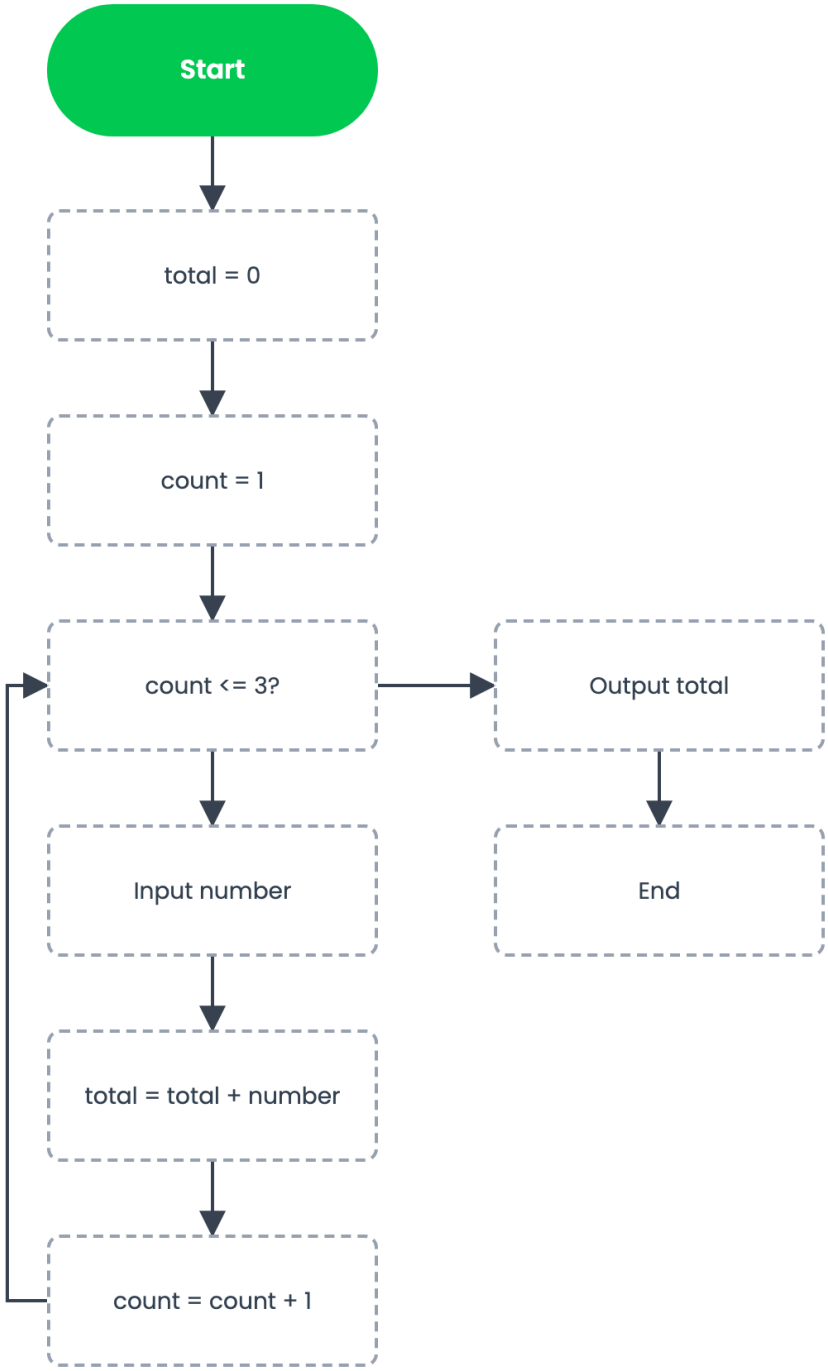


Level 14: Running Total

Medium · Draw Shapes

Input three numbers and output their total. This one combines a counter, a total and a loop.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



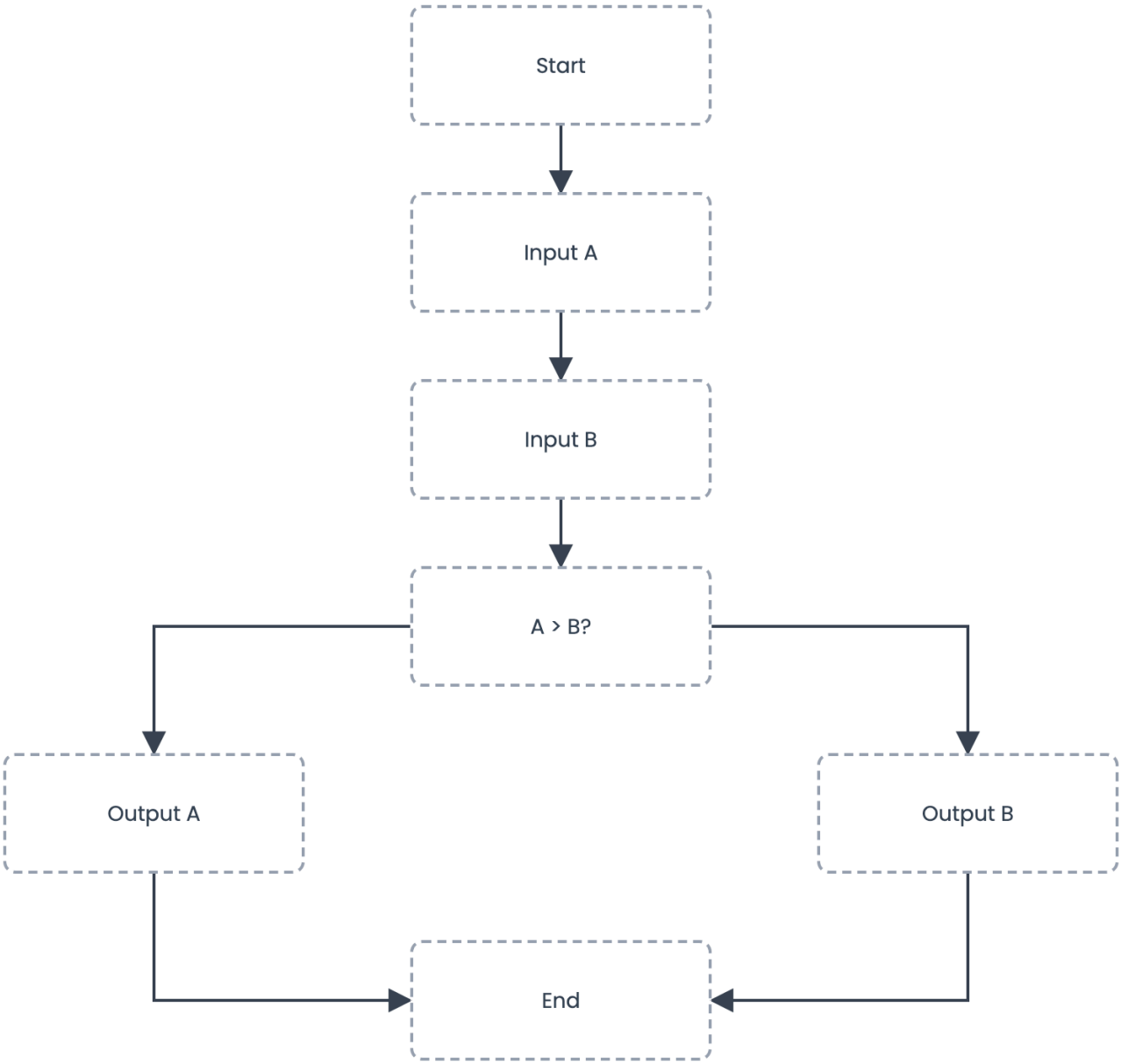


Level 15: Larger Number

Medium · Draw Shapes

Input two numbers and output the larger one.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



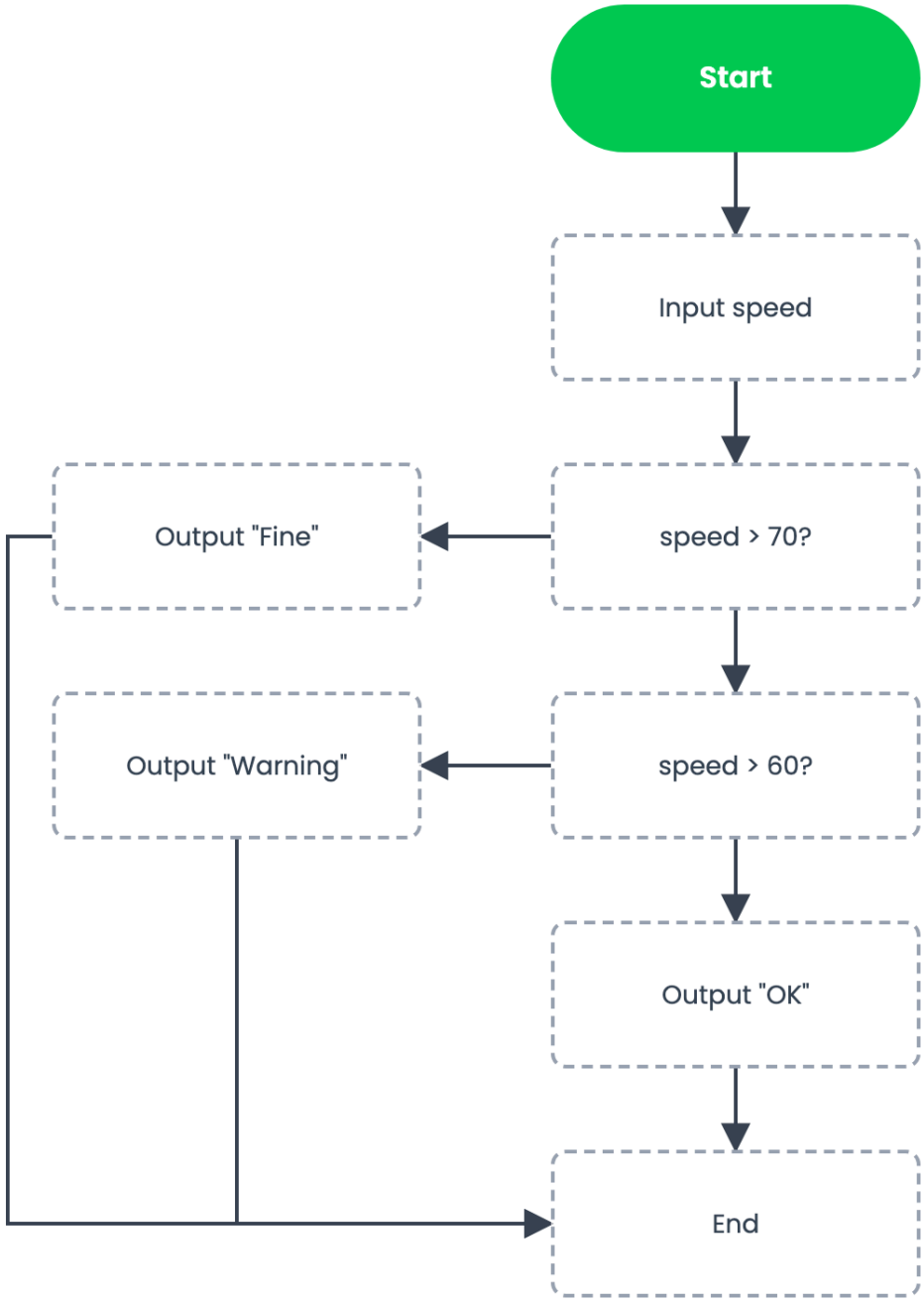


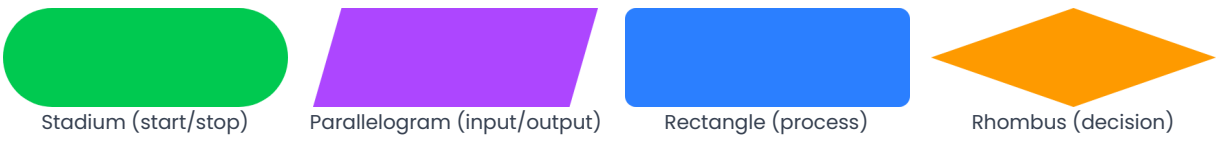
Level 16: Speed Camera

Medium · Draw Shapes

Over 70 is a fine. Over 60 is a warning. Anything else is fine to drive on. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



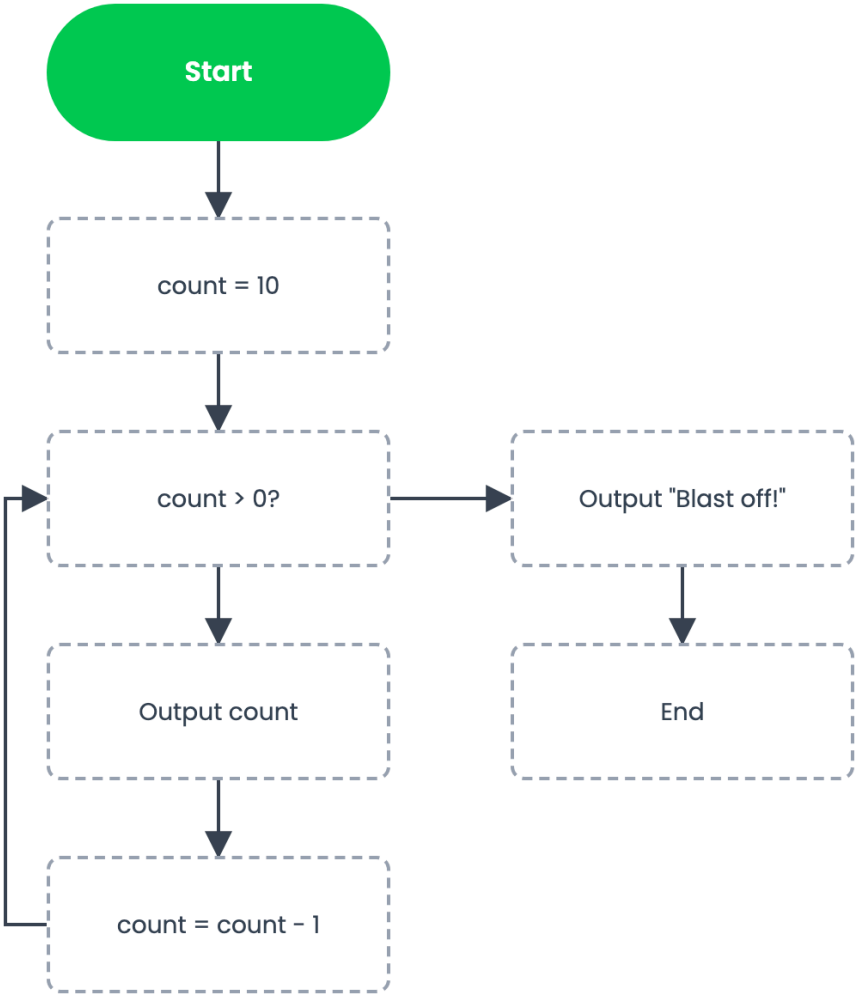


Level 17: Countdown

Medium · Draw Shapes

Count down from 10 and finish with a launch message. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



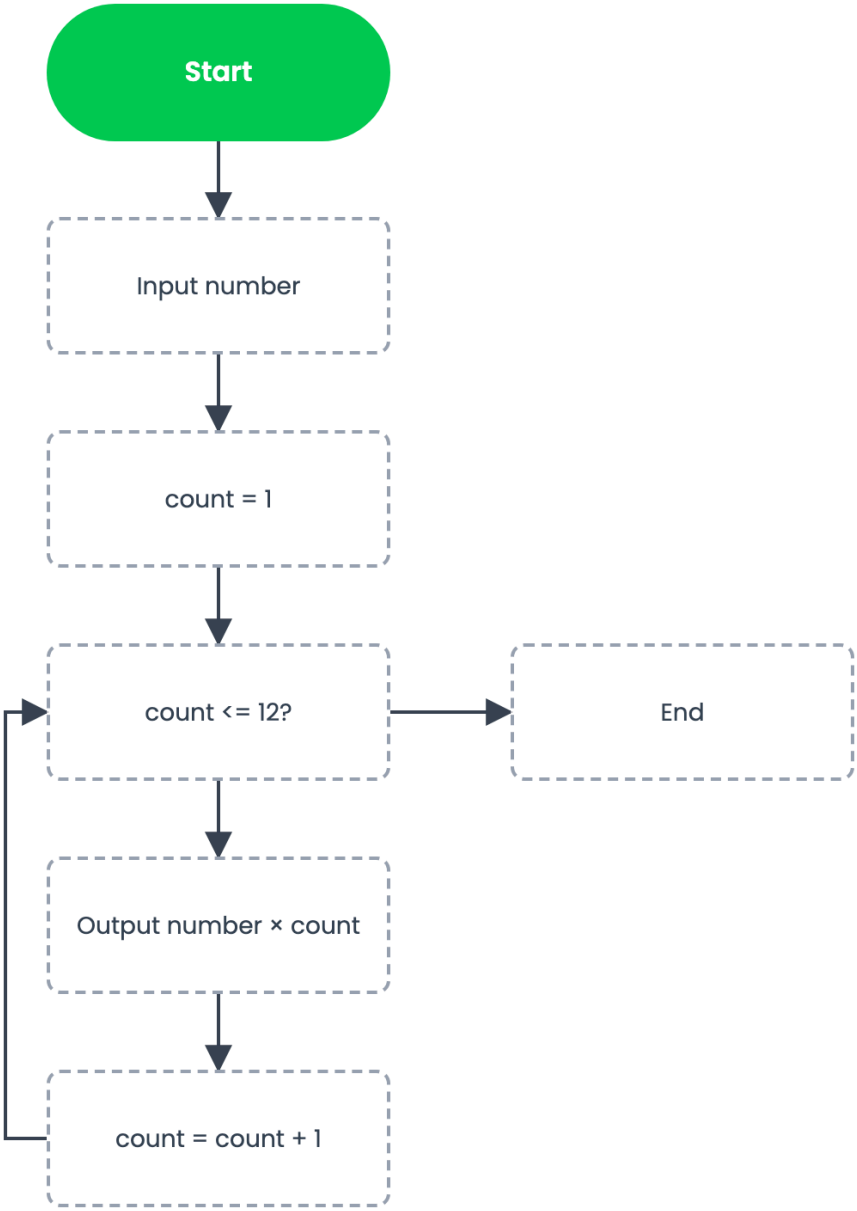


Level 18: Times Table

Medium · Draw Shapes

Input a number and output its 12 times table. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



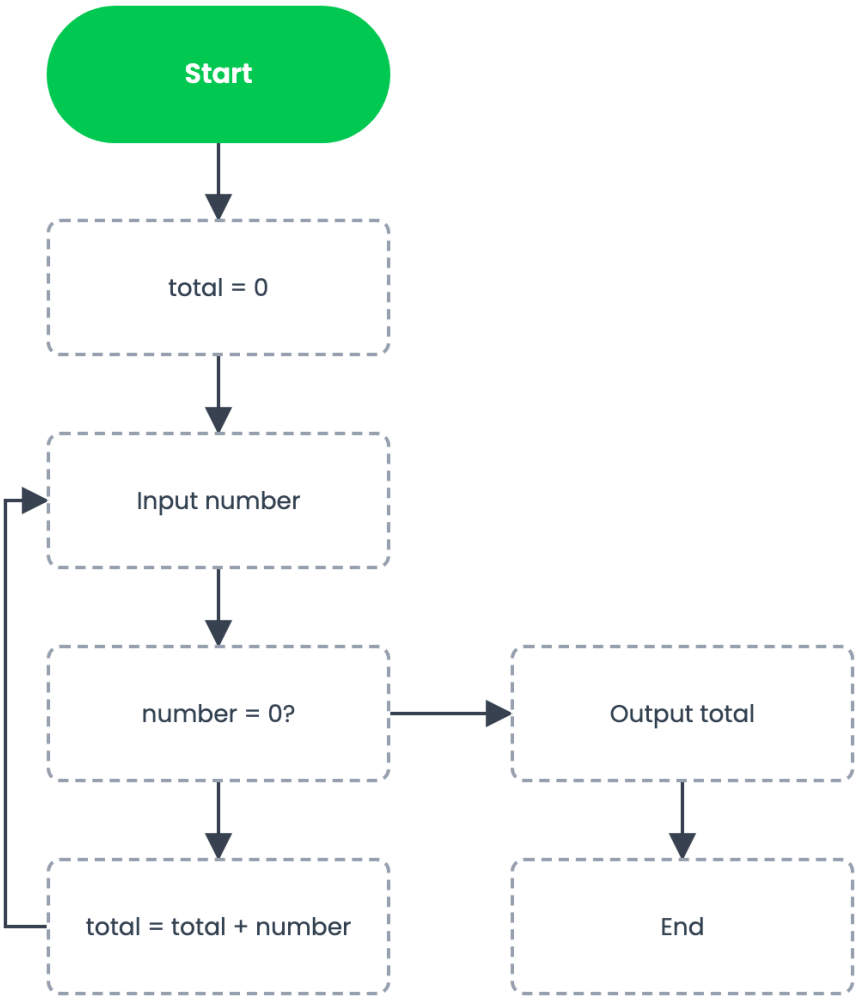


Level 19: Total Until Zero

Medium · Draw Shapes

Keep adding numbers to a total until the user enters 0, then output the total. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



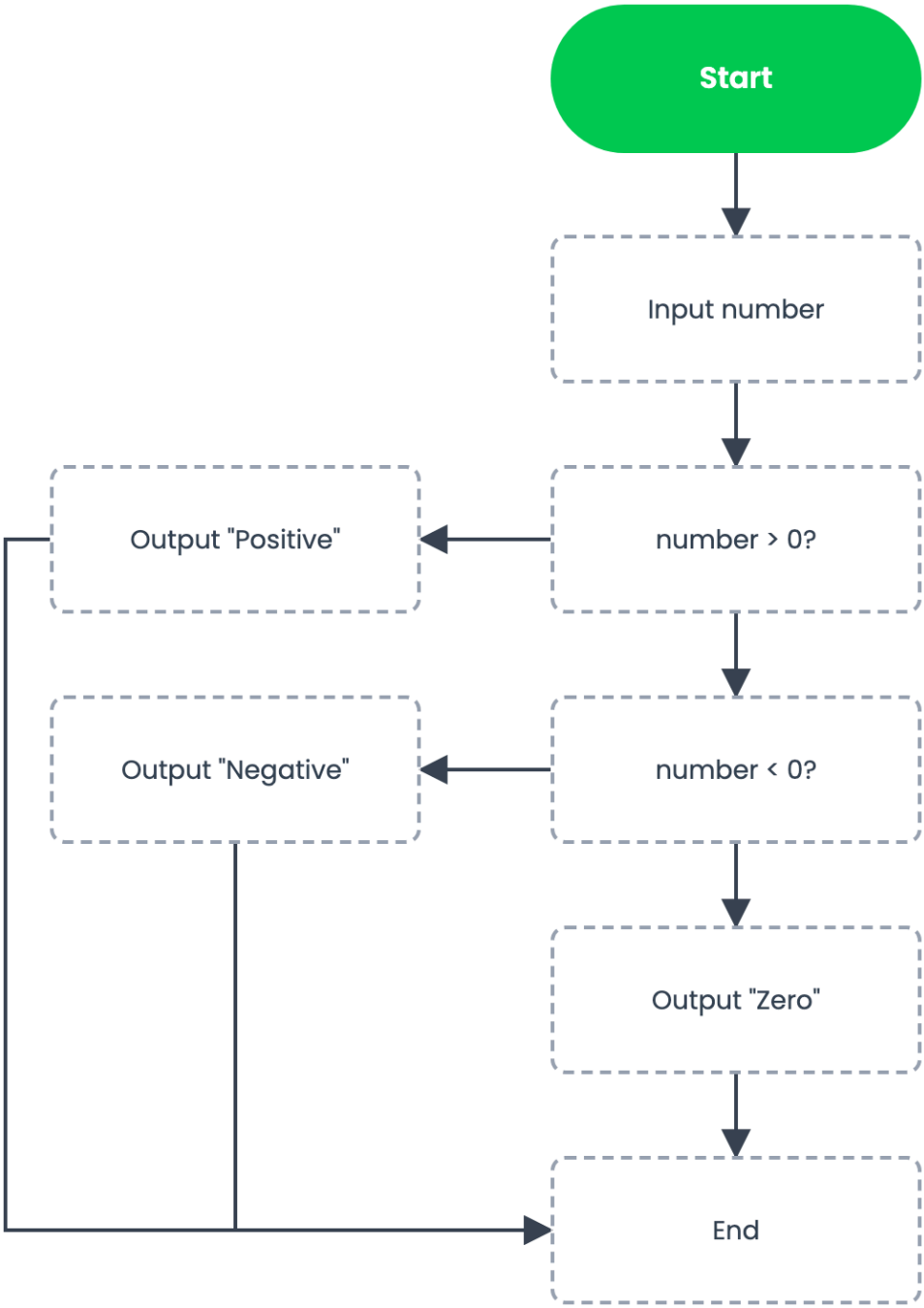


Level 20: Number Sign

Medium · Draw Shapes

Output whether a number is positive, negative or zero. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



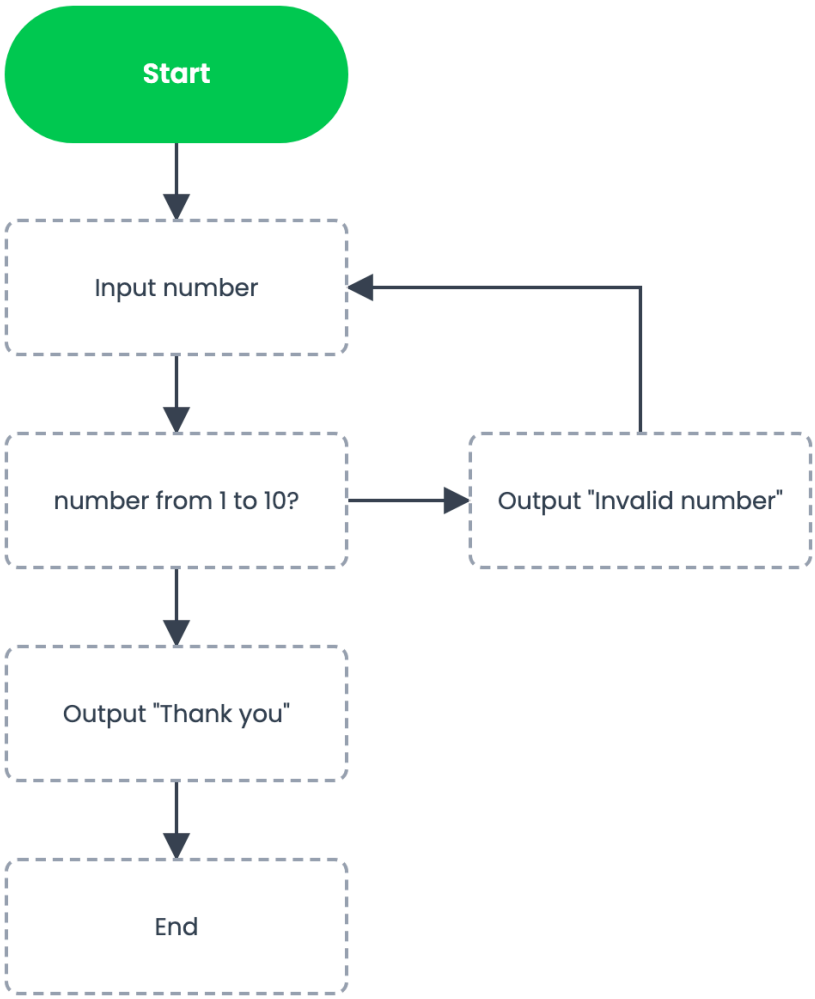


Level 21: Input Validation

Hard · Draw Shapes

Ask for a number from 1 to 10. If it is not in range, say so and ask again. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



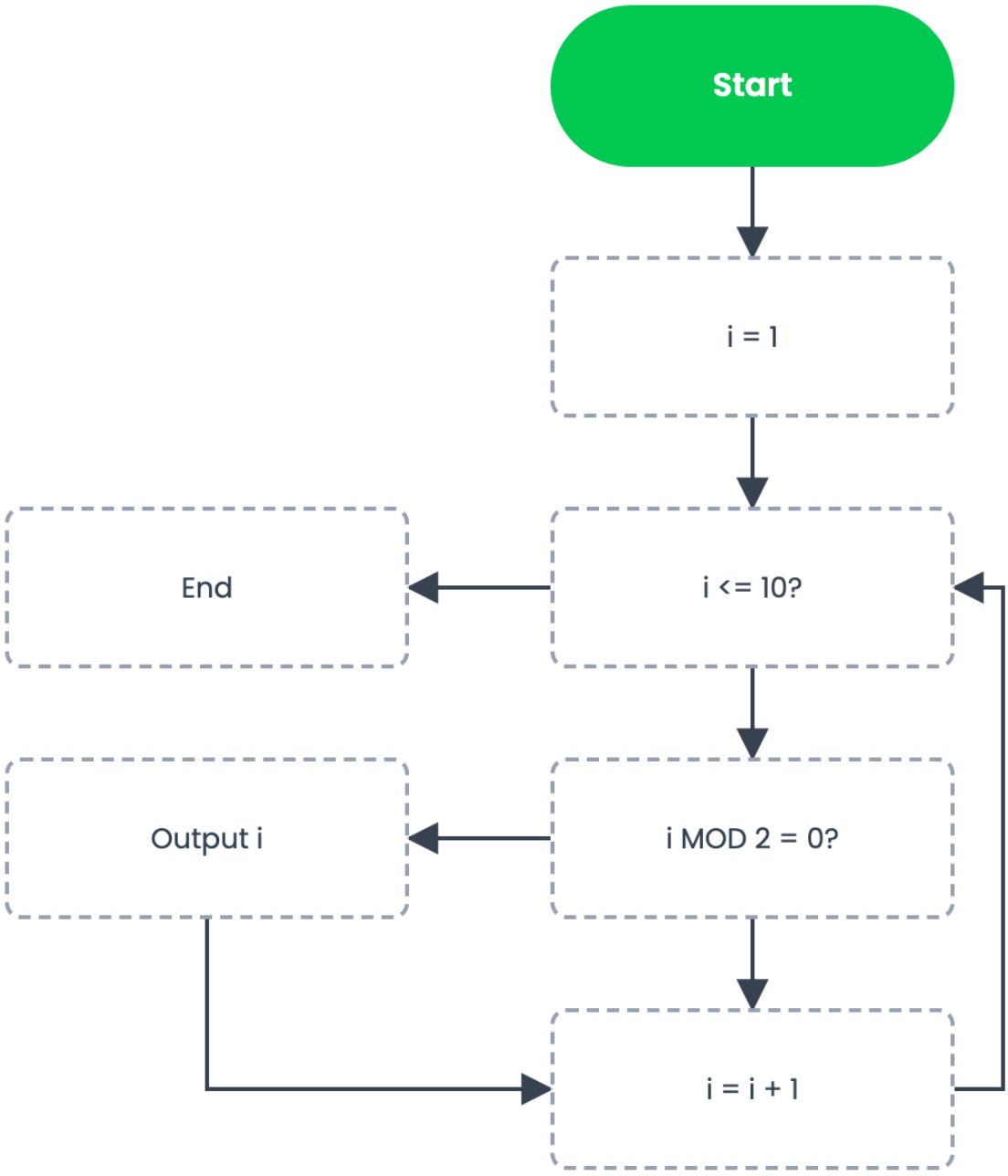


Level 22: Even Numbers

Hard · Draw Shapes

Output only the even numbers from 1 to 10. A decision sits inside the loop. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



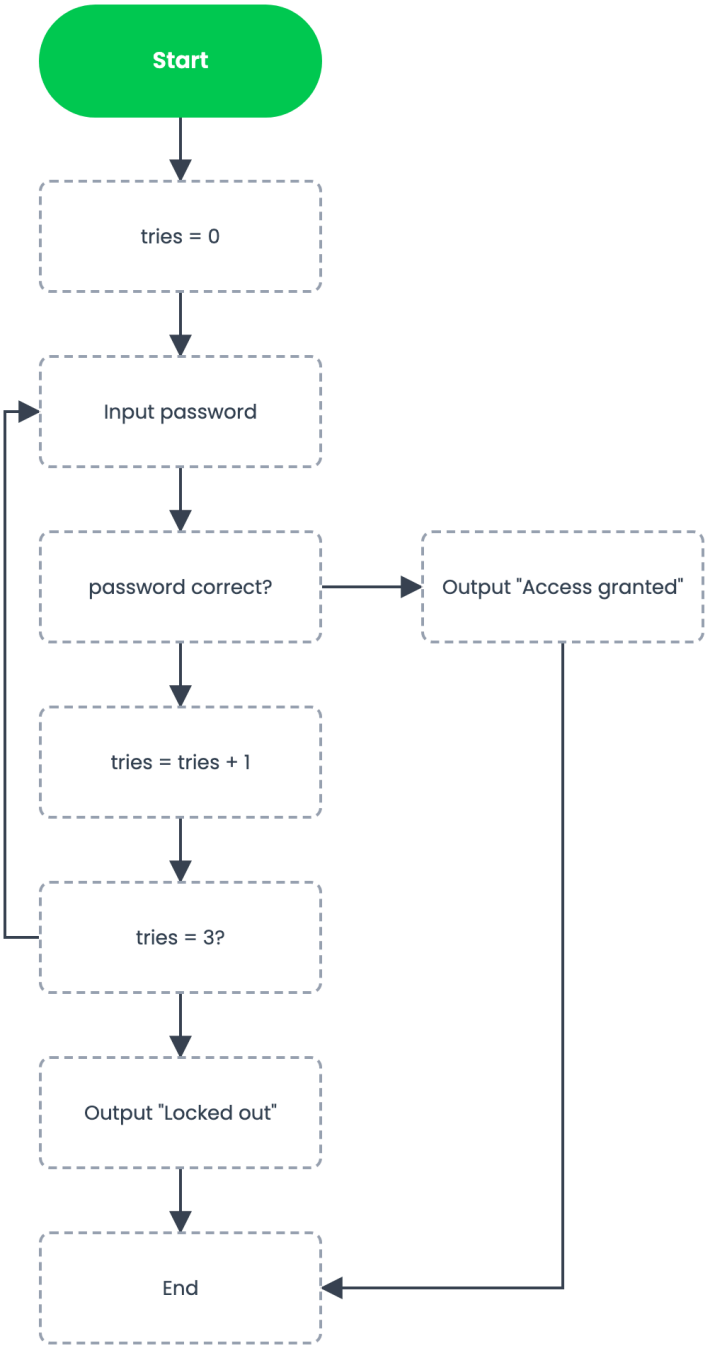


Level 23: Three Attempts

Hard · Draw Shapes

The user gets three tries to enter the correct password. After three wrong tries they are locked out. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



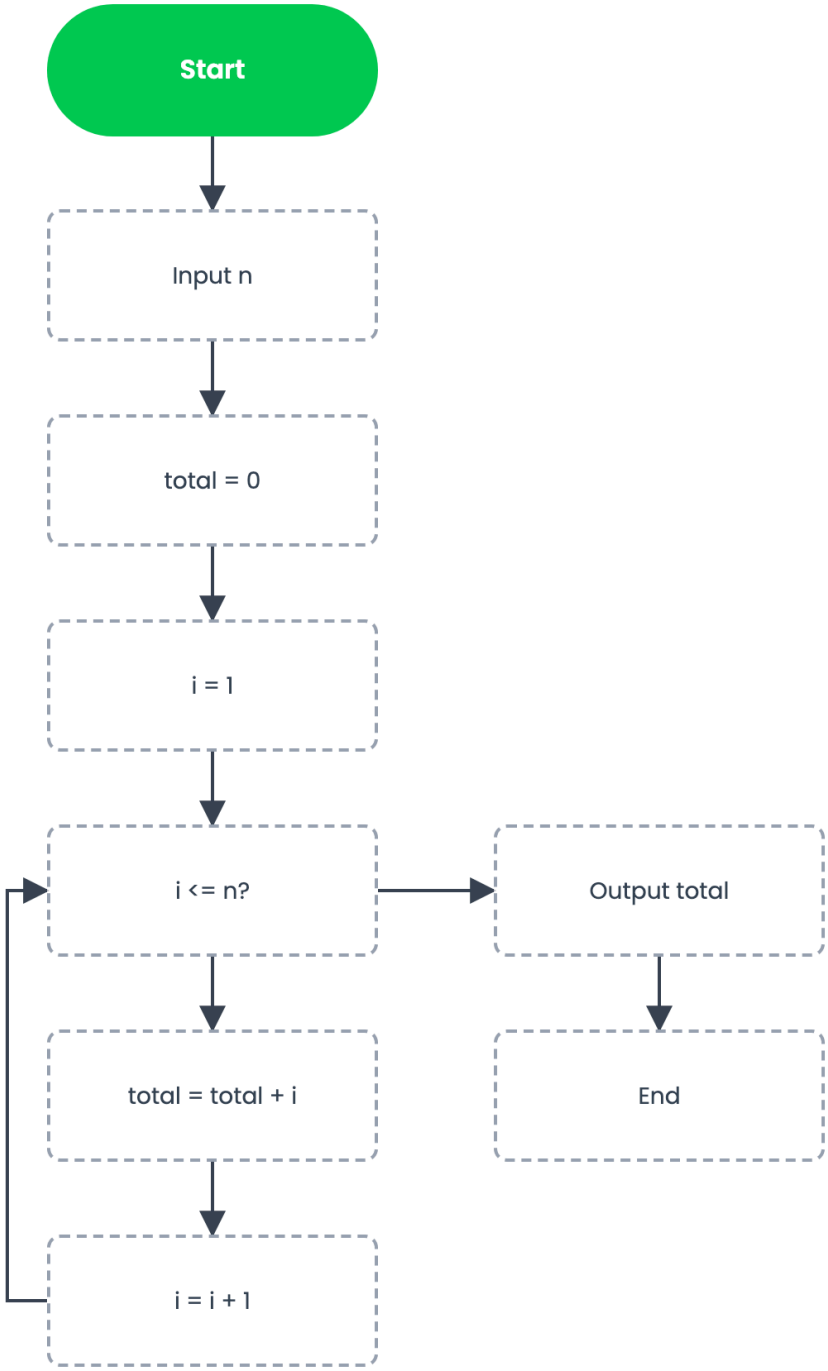


Level 24: Sum to N

Hard · Draw Shapes

Input a number n and output the total of 1 + 2 + … + n. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



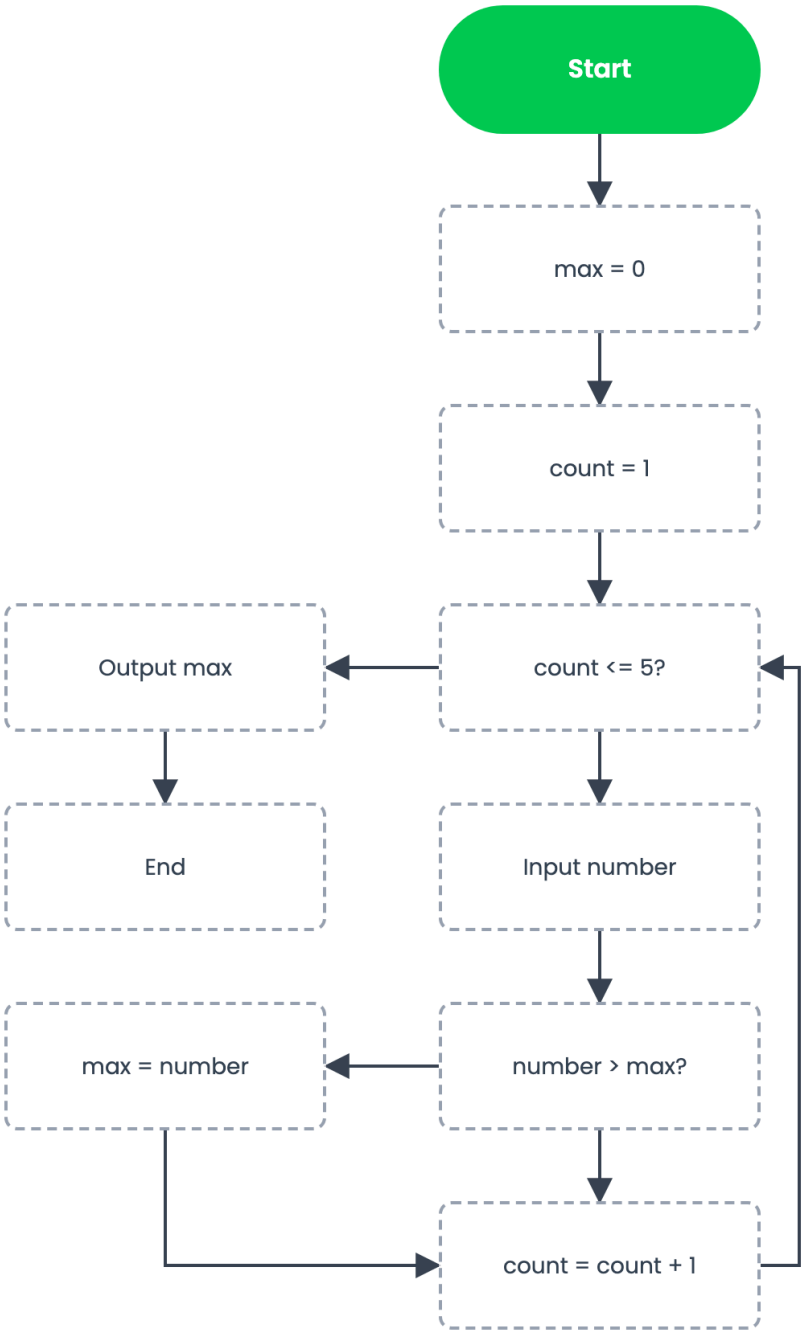


Level 25: Largest of Five

Hard · Draw Shapes

Input five numbers and output the largest. Keep track of the biggest so far. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



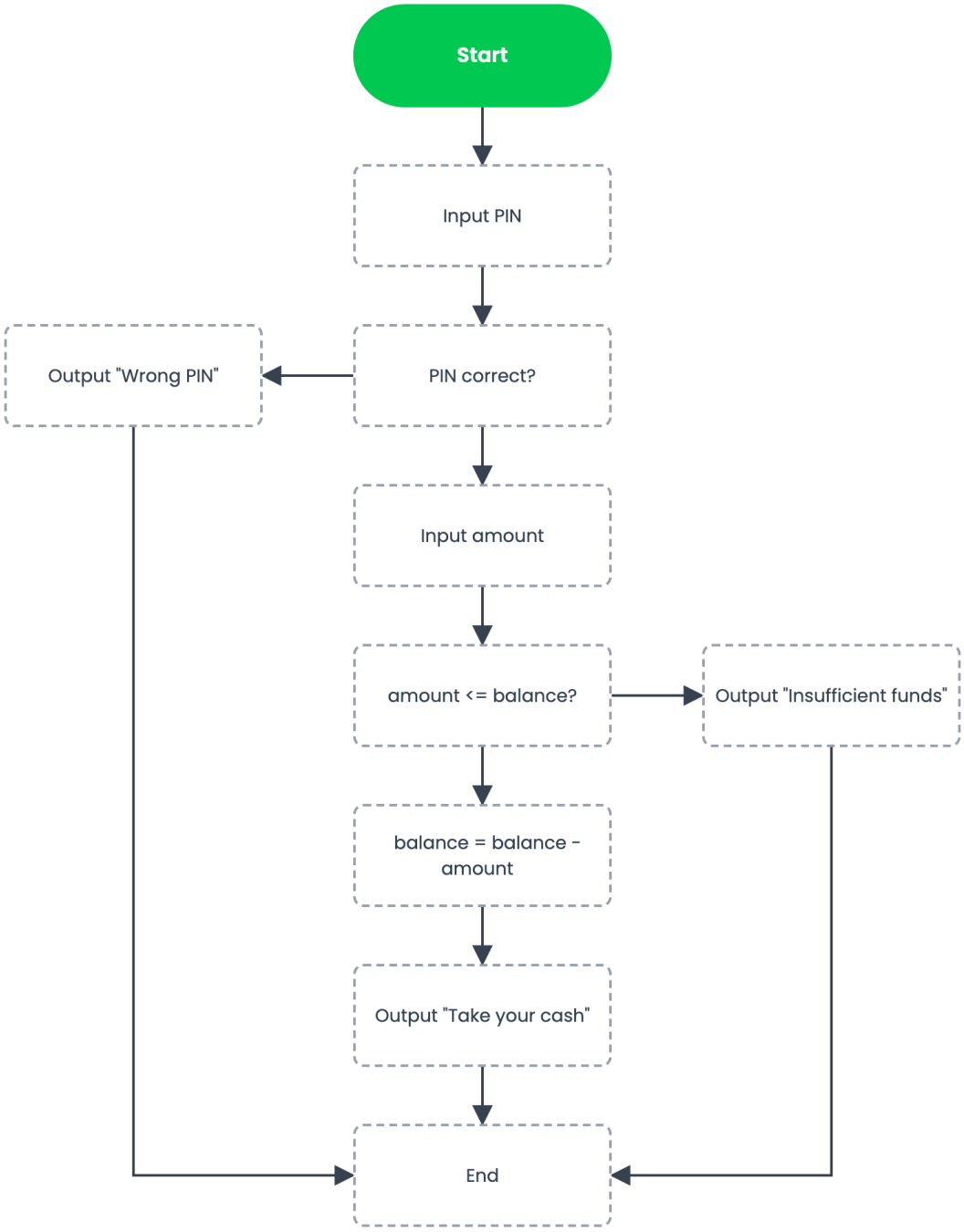


Level 26: Cash Machine

Hard · Draw Shapes

Check the PIN, then check there is enough money before paying out. Every path finishes at End. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



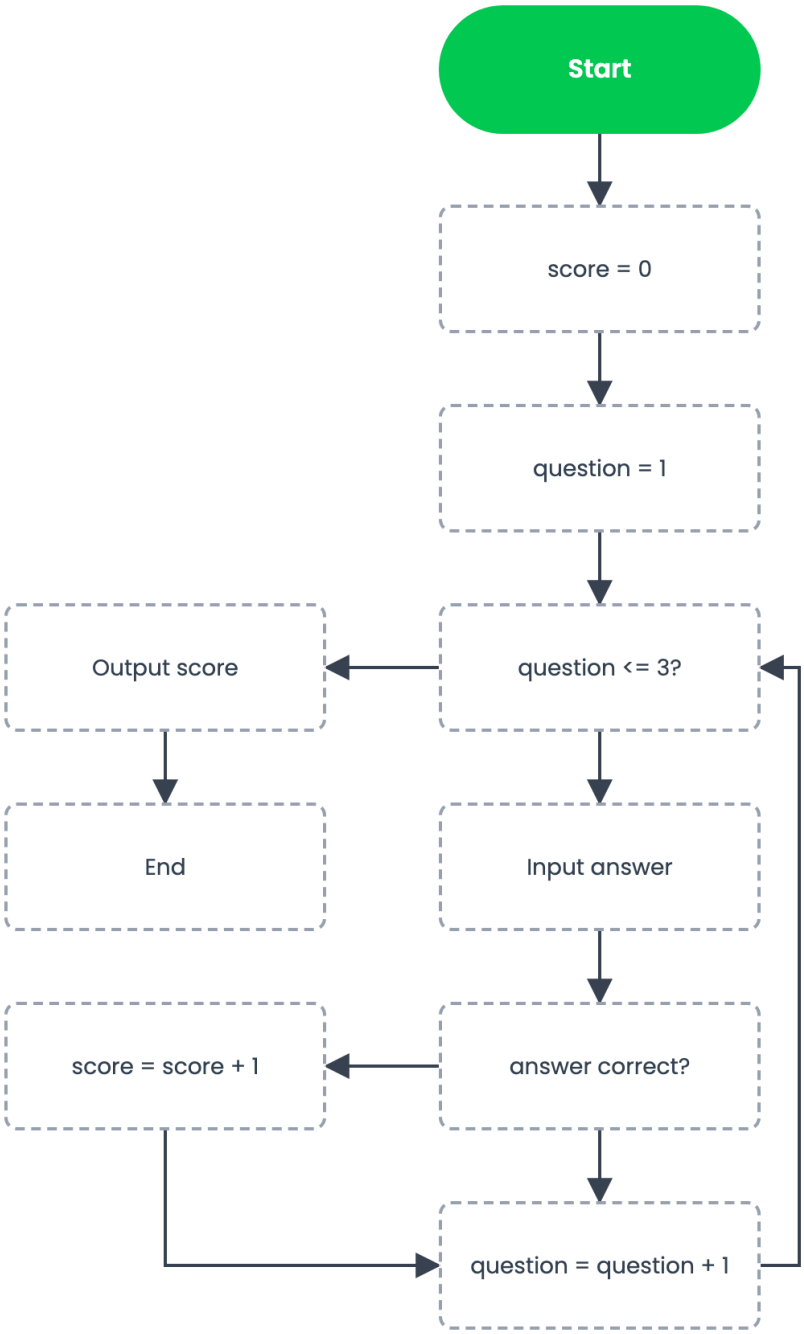


Level 27: Quiz Score

Hard · Draw Shapes

Ask three questions. Add one to the score for each correct answer, then output the score. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



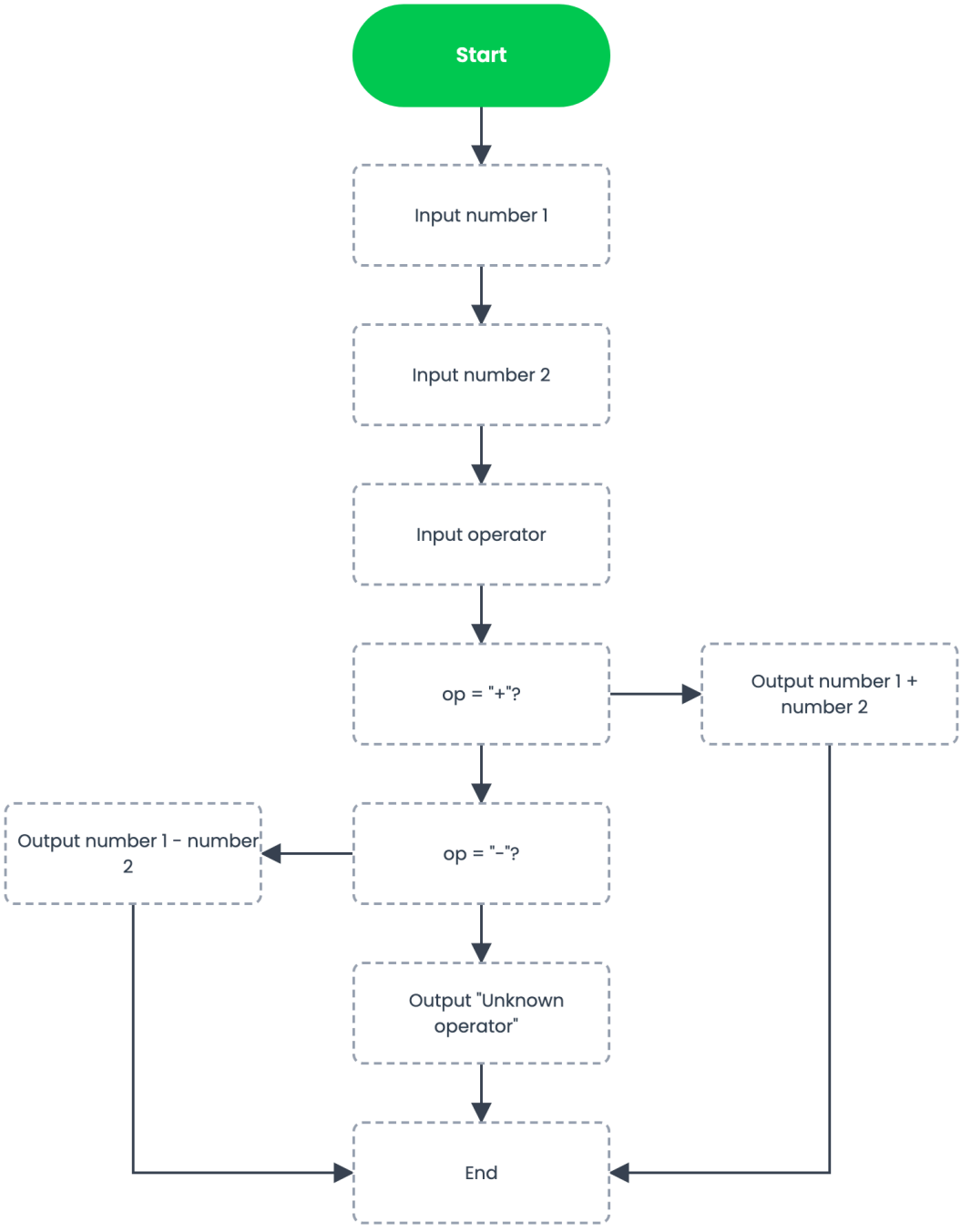


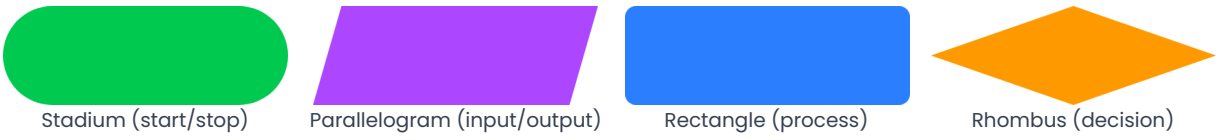
Level 28: Calculator

Hard · Draw Shapes

Input two numbers and an operator (+ or -), then output the answer. Anything else is an unknown operator. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.



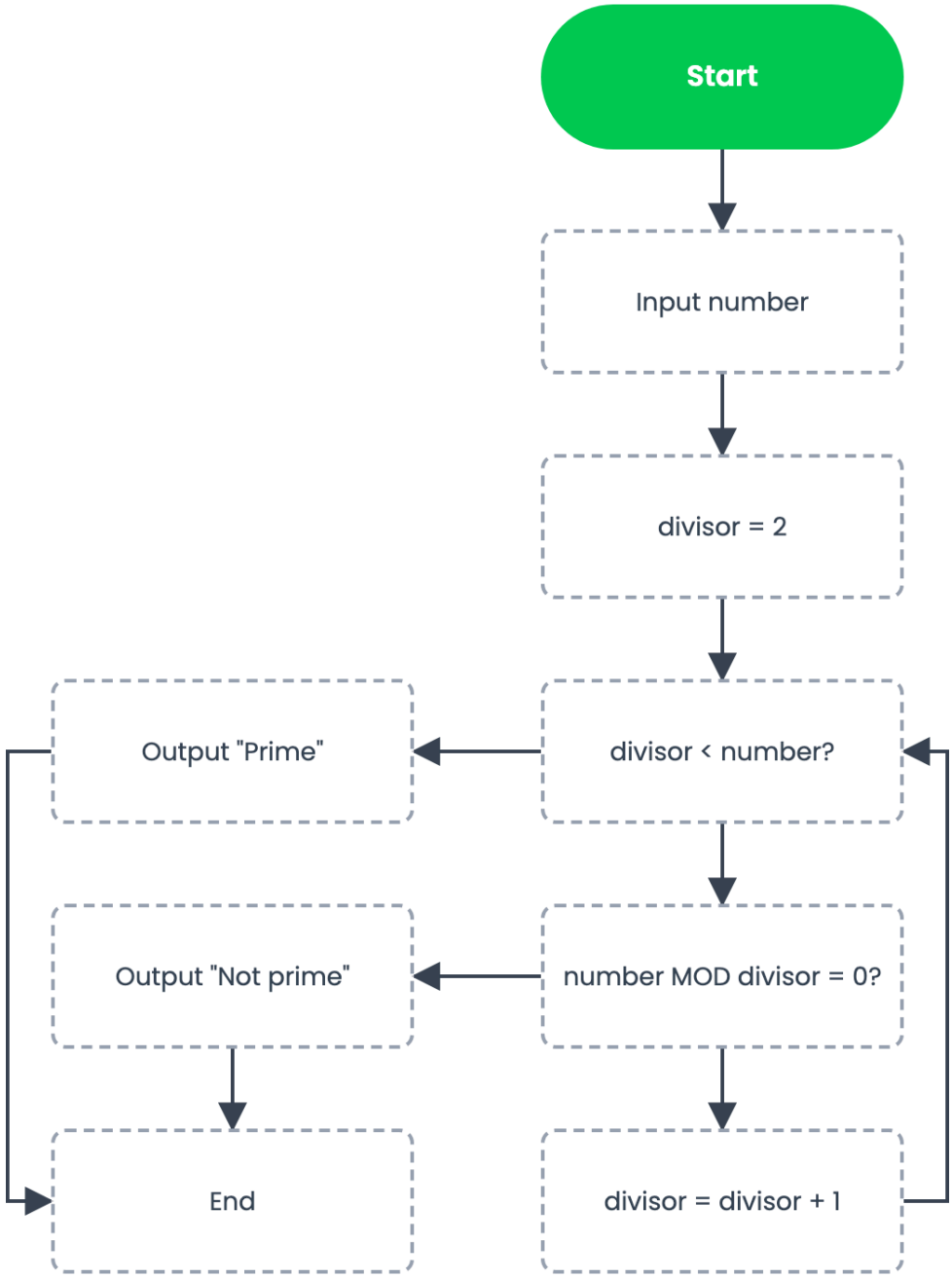


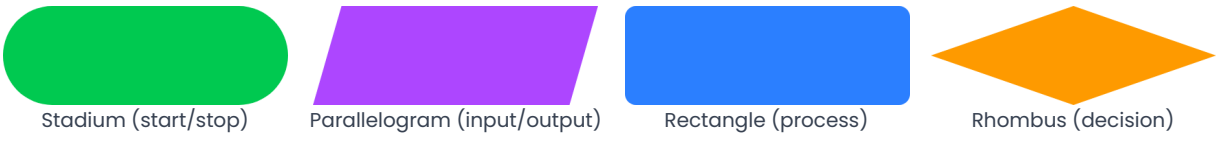
Level 29: Prime Number

Hard · Draw Shapes

Try dividing by 2, 3, 4... up to one less than the number. If any divide exactly it is not prime. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.





Level 30: Menu Loop

Hard · Draw Shapes

Show a menu: 1 says hello, 2 says goodbye, 3 quits. Any other choice asks again. Start is already placed for you.

The text and arrows are already drawn. Draw the correct flowchart shape around the text in each box, and write Yes or No on each arrow leaving a decision.

